Machine Learning for Survival Prediction in Breast Cancer using TCGA gene expression data

BS 845 Data Science and Statistical Modeling in R

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1. Introduction

Breast cancer is one of the most prevalent cancers worldwide, and predicting patient survival remains a key clinical challenge. Advances in RNA sequencing (RNA-seq) enable measurement of gene expression across thousands of genes, providing opportunities for data-driven survival prediction. However, gene expression data are high-dimensional (p >> n), making traditional methods prone to overfitting.

In this study, we use TCGA-BRCA gene expression and clinical data to predict 5-year survival outcomes, defined as a binary variable (survived ≥5 years vs died <5 years). This formulation aligns with standard clinical benchmarks and enables classification modeling.

We compare ridge regression, LASSO, and random forest models, and evaluate their performance using AUC. In addition, we examine feature importance to identify genes associated with survival and assess consistency across methods.

2. Methods

2.1 Data & Preprocessing

The TCGA-BRCA dataset was used, containing RNA-seq gene expression and clinical data for breast cancer patients. After filtering for complete survival information and sufficient follow-up, 351 patients were retained. Patients who were alive but followed for less than five years were excluded to avoid ambiguous outcome labeling.

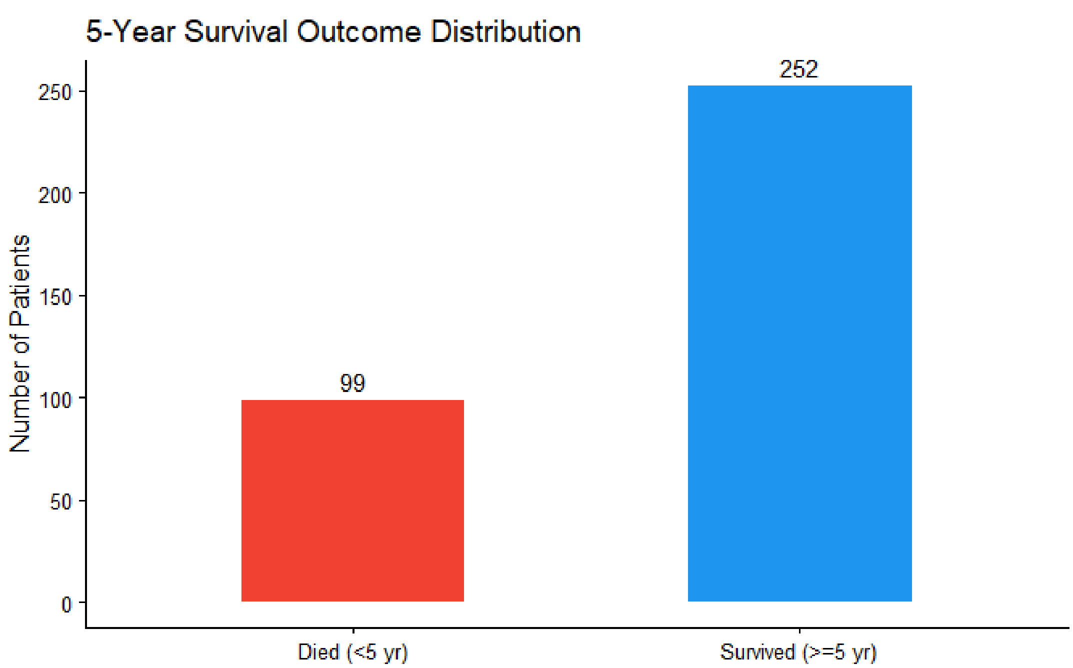

Figure 1. Class distribution of the 5-year survival outcome in the filtered TCGA-BRCA cohort.

The response variable was defined as 5-year survival (1 = survived ≥5 years, 0 = died <5 years), providing a clinically meaningful binary outcome (**Figure 1**). Gene expression values were log-transformed to stabilize variance. Due to high dimensionality (p >> n), feature reduction was performed by selecting the 500 most variable genes (**Figure 2**).

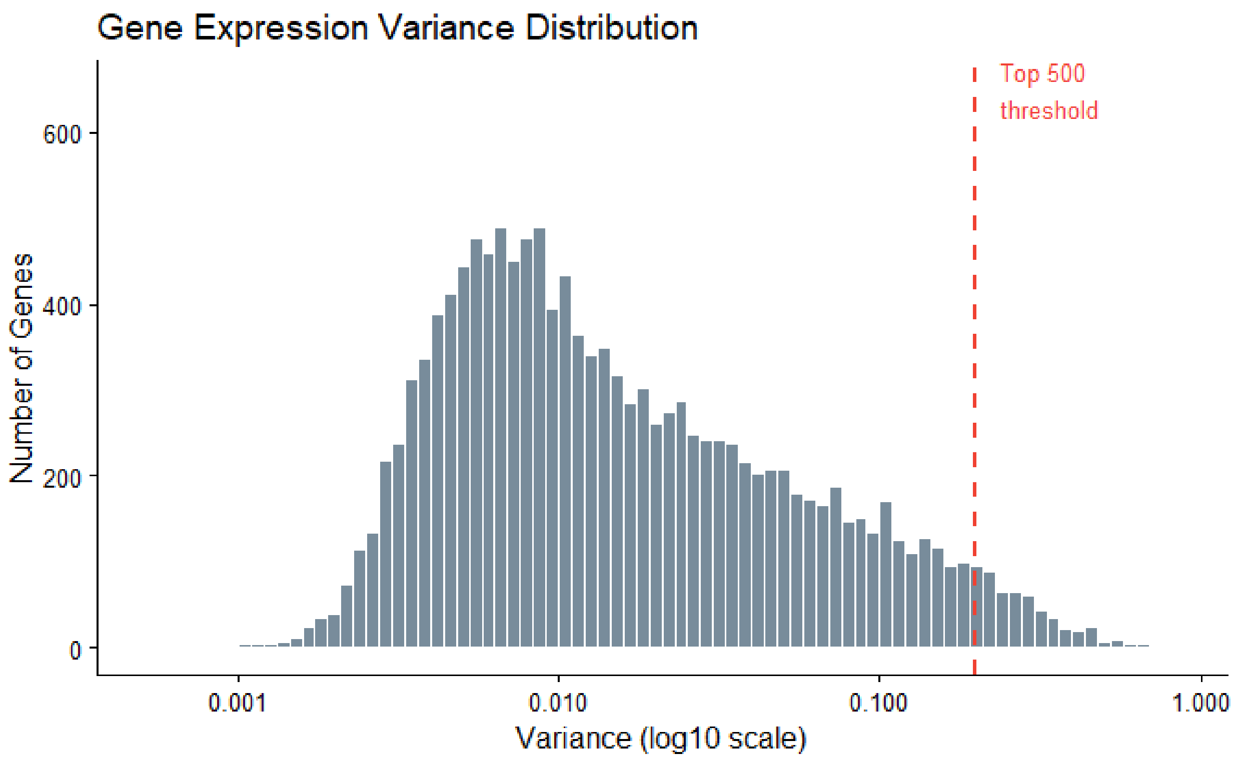

Figure 2. Distribution of per-gene variance (log10 scale). The red dashed line marks the minimum variance among the top 500 selected genes. The majority of genes fall below this threshold and are excluded.

Data were split into training (64%), test (16%), and validation (20%) sets using stratified sampling. The validation set was held out during model development to provide an unbiased estimate of final performance.

2.2 Modeling Approach

Three models were evaluated: ridge regression, LASSO, and random forest. Ridge regression applies L2 regularization to shrink coefficients and reduce overfitting while retaining all features. LASSO applies L1 regularization, performing feature selection by setting some coefficients to zero. Random forest, a tree-based ensemble method, captures nonlinear relationships and interactions among genes.

2.3 Model Training and Evaluation

Hyperparameters were tuned using 5-fold cross-validation on the training set. Models were compared on the test set, and the best-performing model was evaluated on the held-out validation set.

Performance was assessed using the area under the ROC curve (AUC), which is appropriate for imbalanced data and measures the model's ability to distinguish between classes across all thresholds.

3. Methods

The TCGA-BRCA dataset was used, with 351 patients retained after filtering for complete survival information and sufficient follow-up. The outcome was defined as 5-year survival (1 = survived, 0 = died). Gene expression values were log-transformed, and the top 500 most variable genes were selected to address high dimensionality ($p >> n$).

Data were split into training (64%), test (16%), and validation (20%) sets using stratified sampling. Three models were evaluated: ridge regression, LASSO, and random forest. Hyperparameters were tuned using 5-fold cross-validation on the training set. Model performance was assessed using AUC, which is appropriate for imbalanced data.

4. Results

Ridge achieved the highest test performance (AUC = 0.633), followed by random forest (0.610) and LASSO (0.585) **(Figure 3)**. On the validation set, ridge achieved an AUC of 0.727 **(Figure 4)**, indicating reasonable generalization.

LASSO selected 70 genes, while random forest identified important features based on permutation importance (Figure 5). Two genes (GLT25D2 and BMP7) were identified by both methods, suggesting more robust signals. Overall, predictive performance is modest.

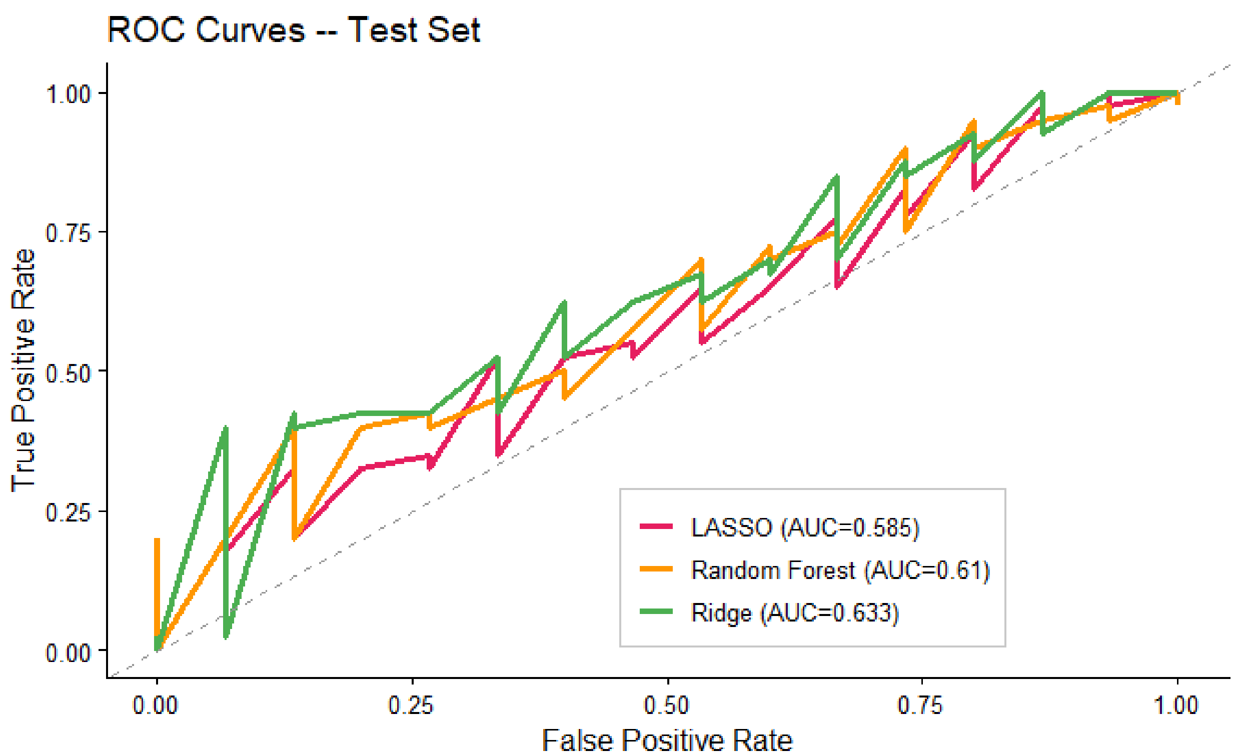

Figure 3. ROC curves for ridge, LASSO, and random forest models on the test set.

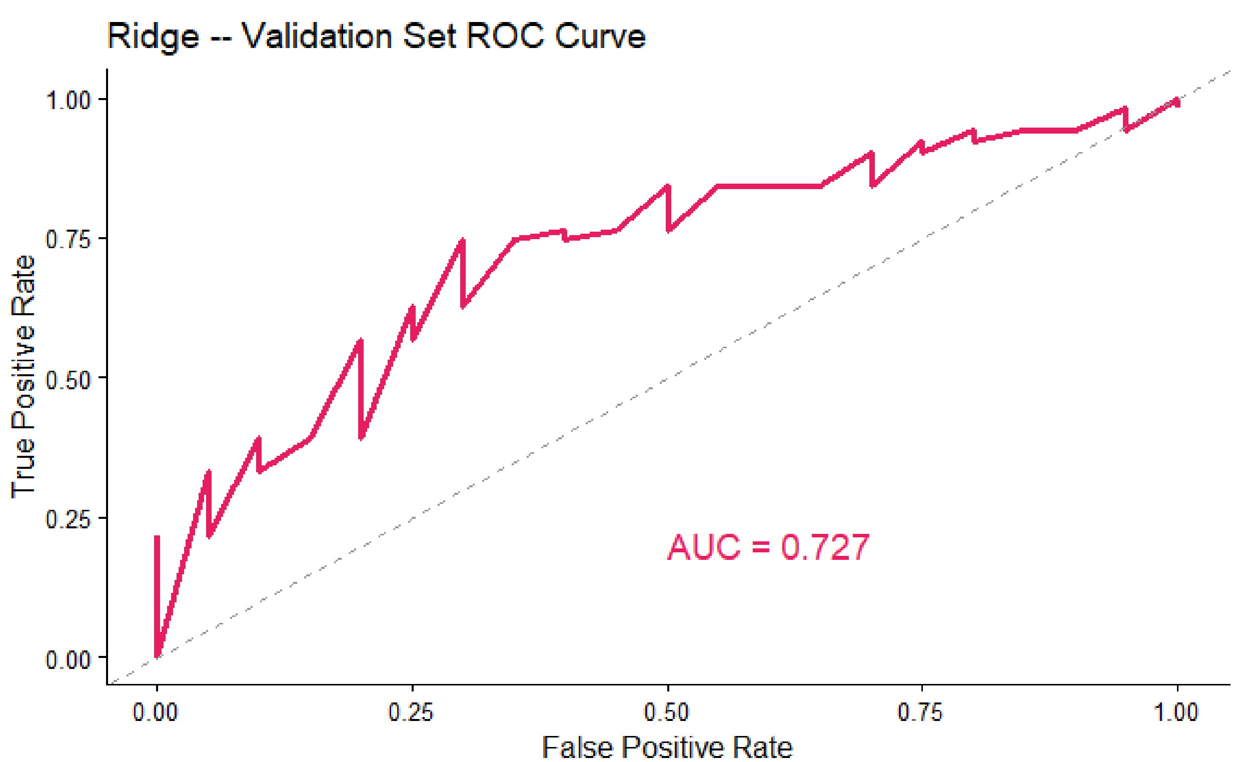

Figure 4. ROC curve for ridge regression on the validation set.

5. Discussion

While ridge performed best, overall predictive accuracy remains limited, reflecting the complexity of survival outcomes and high-dimensional gene expression data. Different models capture complementary aspects of the data, with LASSO providing feature selection and random forest capturing nonlinear effects.

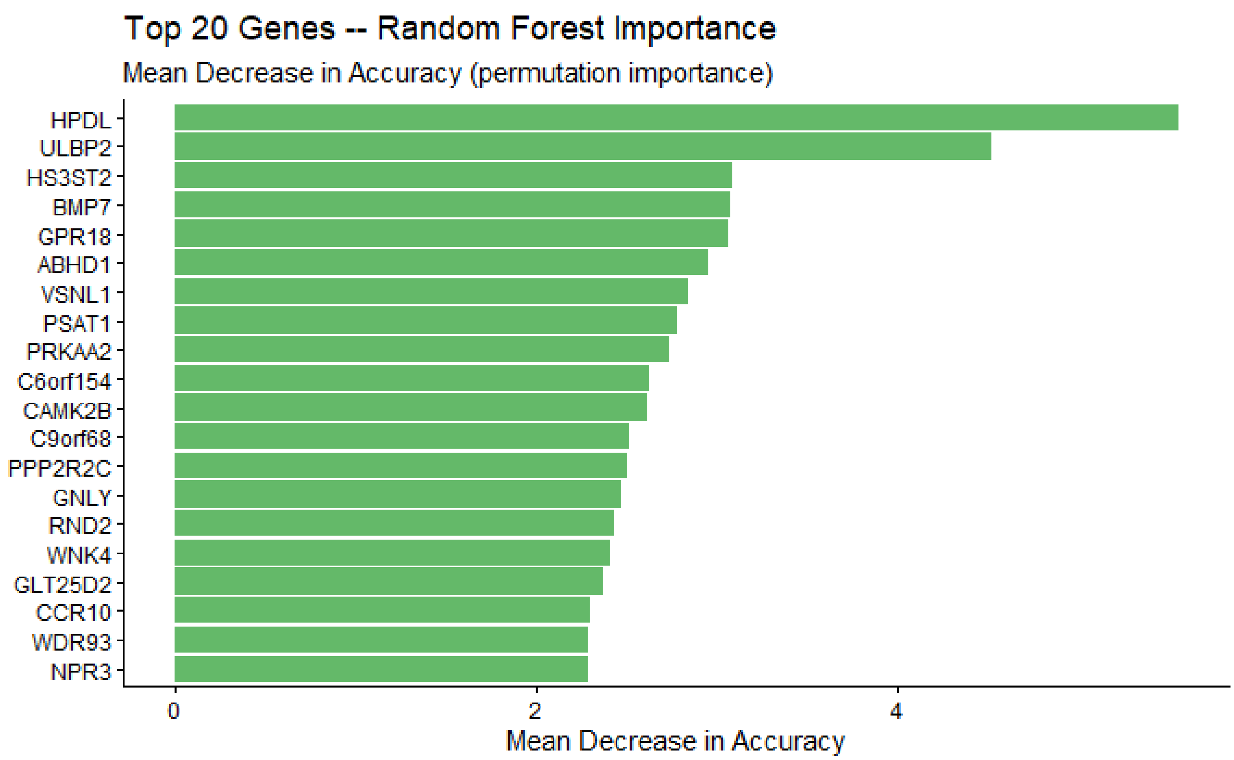

Figure 5. Top 20 genes ranked by Random Forest feature importance.

Limitations include small sample size and class imbalance. Future work could incorporate additional data types and more advanced modeling approaches to improve performance.